D2 Detailed Design

Project: HepatoFusion

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Document: Design D2 (Week 8). Sits beside D0 and D1. Does not overwrite them.

AI use. An AI tool helped draft this document and the figures. The partial-score rule change, the identifier-gate deidentification approach, the modality-integrator contract for pre-annotated DICOM, the edge-case decisions, the traceability mapping, and the rationale are the team's decisions. The tool wrote the structure; the team made the calls.

Team decisions used in this document

Topic	Team decision
D2 scope	Finish Identifier gate, Radiology intake, Genomics intake, Modality integrator, Risk scorer, Review package + glue (Auth, Notifications, Config, Logging).
Partial-score rule	Score only when pathology + cleaned outcomes + at least one of {radiology, molecular}. Otherwise insufficient_modalities; scorer does not run. Replaces D0 D-06.
Identifier gate	Deidentify per CCHMC instructions; block if identifier cannot be removed (e.g. burned-in pixels). Field list = OQ-01 config.
Risk scorer model	Interface specified; model TBD (OQ-04). Swappable callable.
Integrator radiology	DICOM counts only when annotated; no in-pipeline raw DICOM feature extractor.
PathPresenter	Account access yes; import format = OQ-02.
Glue Auth / Notify / Config / Log	SSO folder ACLs; no push notifications; pipeline_config.yaml; Python logging under v1/logs/.
D0 / D1 text	Not modified; no peer-review errata at time of writing.

1. Header, scope, and conventions

Title: HepatoFusion

Goal (same wording as D0 and D1): Put an infrastructure in place for radiology, molecular data, and cleaned clinical outcomes on the existing pediatric liver-cancer cohort, then integrate those modalities with the finished pathology localization so a pathologist can use a research risk score as a helper. The finished pathology U-Net is also planned to surface inside PathPresenter (QuPath is the alternative viewer still under consideration), so the localization is available in the systems pathologists already use.

Scope of D2. The six D0 components deferred by D1 are detailed here with data model (or no persistent data), behavior, and interface. Four glue components (Authentication, Notifications, Configuration, Logging) are also detailed. D2 also delivers the edge-case table, requirements traceability, design rationale, and peer-review notes scaffold.

Conventions: solid = we build / entity; dashed = external; arrow = file handoff (I1–I18); red = error path; study_id is the cohort key (no PHI attributes).

Two-minute overview. HepatoFusion is a file pipeline that collects radiology, molecular, and clinical outcomes on an existing 102+ patient cohort whose pathology U-Net already produces tumor maps. Files are deidentified, indexed by study_id, integrated when enough modalities exist, scored, and handed to PathPresenter. D1 covered the data model and outcomes cleaner. D2 covers everything else. Not in D2: final model choice, final identifier-field list from CCHMC, PathPresenter import format.

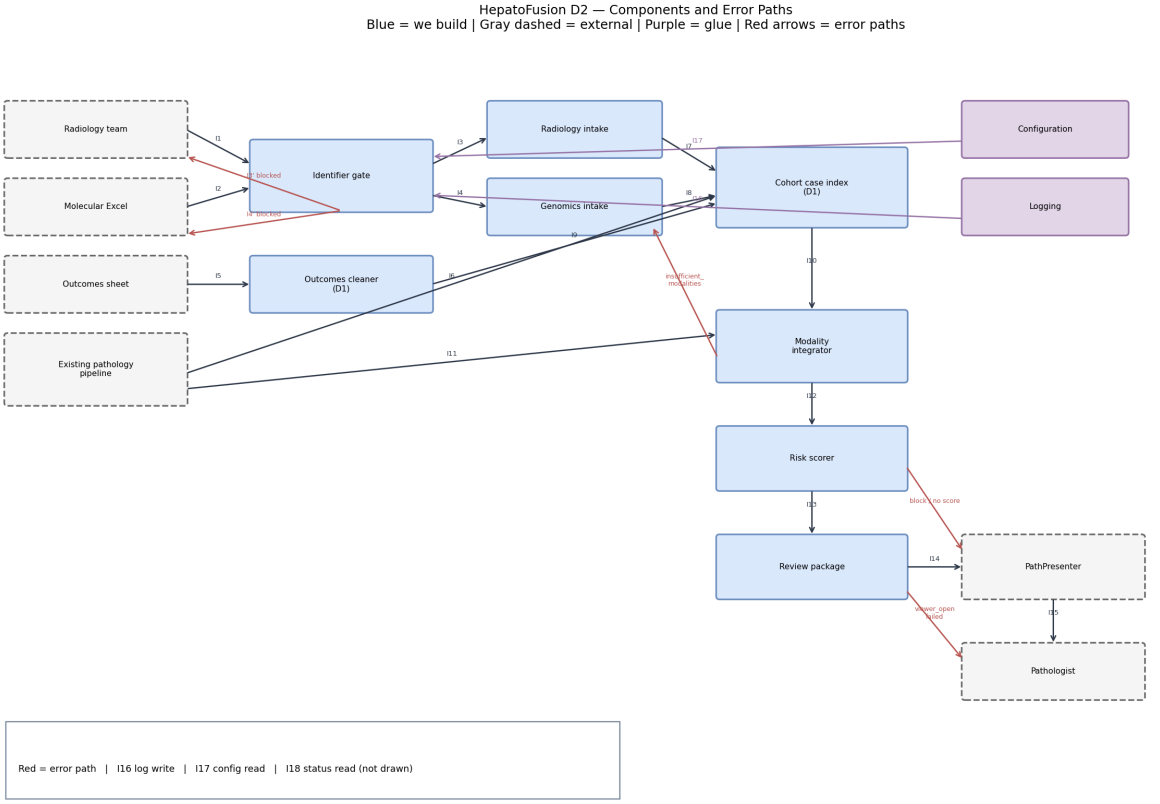
Open questions

ID	Open question	Owner	Why still open
OQ-01	Exact identifier-field list for the gate	Shivam	Awaiting CCHMC written guidance; rules in config.
OQ-02	PathPresenter import format for overlay + helper	Shivam + IT	Access live; format under discussion.
OQ-03	Log retention / rotation	Nipun	Share quota / institutional rules; stub 90 days.
OQ-04	Final risk-scorer model	Both	Holdout comparison running; interface fixed.
OQ-05	Whether QuPath becomes wired if PathPresenter blocks	Both	Deferred until OQ-02.

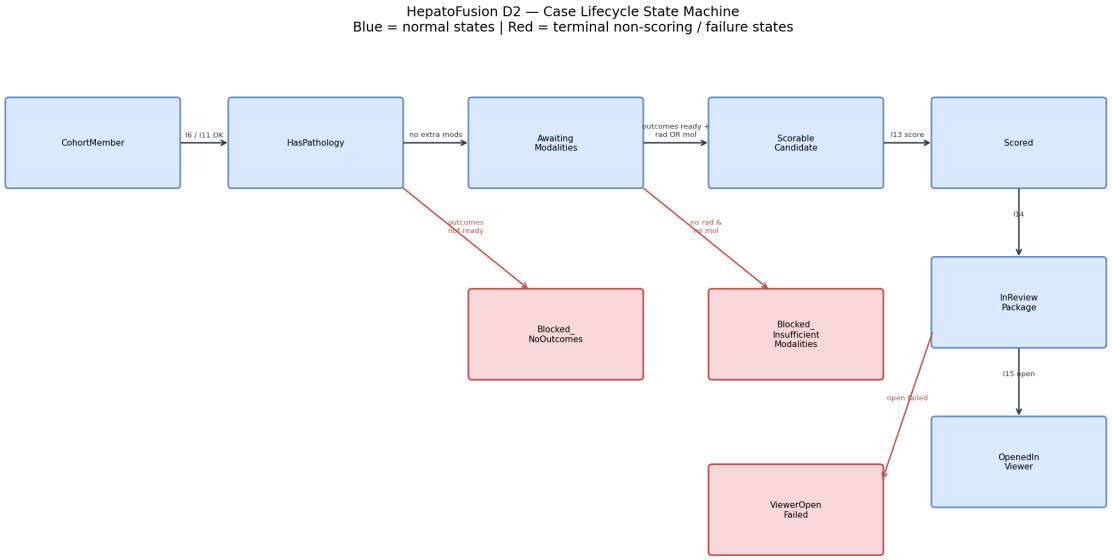
2. Finished components (D2 diagrams and specs)

2.1 Overview — components and error paths

Red arrows mark error paths. Editable source: D2_diagrams.drawio.



Case lifecycle state machine (red = terminal non-scoring / failure)



2.2 Identifier gate (US-05, I1/I2)

(a) Data model. GateRecord (incoming_path, received_at, study_id, format, decision ∈ {passed, blocked, passed_with_redaction}, …). RedactionEntry (field_type, action, location_hint) — never stores identifier values. Rules from pipeline_config.yaml (OQ-01).

(b) Behavior. Pick up incoming file → validate format → deidentify DICOM tags / xlsx columns → block on burned-in pixel text (E_PIXEL_IDENTIFIER) → write gate_status.json → copy to accepted/ or leave in incoming/ → log via I16.

Name	Inputs	Outputs	Errors
scan_and_deidentify	incoming_path, study_id, format, rule_set_version	gate_status.json + file in accepted/ or blocked	E_UNKNOWN_FORMAT, E_PIXEL_IDENTIFIER, E_REDACTION_FAILED, E_CONFIG_MISSING_RULES, E_IO
status_for	incoming_path	gate_status.json	E_NOT_FOUND

D0 I1/I2 revisited: three-way outcome (passed / passed_with_redaction / blocked) instead of binary pass/block.

2.3 Radiology intake (US-03, I3, I7)

(a) RadiologyRecord in radiology.json: study_id, modality_type CT|MRI, source_path, annotation_status, link_status, received_at. No DICOM conversion.

(b) Watch accepted DICOM → resolve study_id against cohort (linked / unmatched / ambiguous) → check annotation sibling → write radiology.json → emit I7. Integrator enforces annotated-only for scoring; intake records state.

Name	Inputs	Outputs	Errors
register_radiology	study_id, DICOM path, cohort list	radiology.json, I7	E_UNMATCHED_STUDY, E_AMBIGUOUS_STUDY, E_IO, E_SCHEMA
read_radiology	study_id	radiology.json	E_NOT_FOUND, E_SCHEMA

2.4 Genomics intake (US-02, I4, I8)

(a) GenomicsRecord in molecular.json: study_id, format=xlsx, source_path, link_status, received_at. Non-xlsx blocked at gate (D0 D-05).

(b) Same structure as radiology without annotation_status. Emit I8.

Name	Inputs	Outputs	Errors
register_molecular	study_id, xlsx path, cohort list	molecular.json, I8	E_UNMATCHED_STUDY, E_AMBIGUOUS_STUDY, E_IO, E_SCHEMA
read_molecular	study_id	molecular.json	E_NOT_FOUND, E_SCHEMA

2.5 Modality integrator (US-02, I10–I12)

(a) IntegratedInputList + InputRef → input_list.json. assembly_decision ∈ {sufficient, insufficient_modalities, outcomes_not_ready, unmatched}.

(b) Partial-score rule (D2): sufficient only if pathology linked AND outcomes model_ready AND (radiology linked with annotation present OR molecular linked). Radiology without annotation counts as not linked for scoring.

```
has_pathology / has_outcomes / has_radiology(annot) / has_molecular
if not pathology → unmatched
elif not outcomes → outcomes_not_ready
elif not (radiology or molecular) → insufficient_modalities
else → sufficient; write input_list.json
```

Name	Inputs	Outputs	Errors
assemble_input_list	study_id	input_list.json	E_PATHOLOGY_POINTER_MISSING, E_CASE_FOLDER_MISSING, E_SCHEMA, E_IO
read_input_list	study_id	input_list.json	E_NOT_FOUND

2.6 Risk scorer (US-02, I12–I13)

(a) ScoreResult → score_result.json. Model artifact via model_version (OQ-04). No training data on disk.

(b) Read assembly_decision; block on unmatched / outcomes_not_ready / insufficient_modalities; else load model, build features from used_inputs only (no fabricated zeros), predict ∈ [0,1], write result + disclaimer “research prototype, not a diagnostic device”.

Name	Inputs	Outputs	Errors
score_case	study_id, model_version	score_result.json	E_INPUT_LIST_MISSING, E_MODEL_LOAD, E_FEATURE_MISSING, E_MODEL_OUT_OF_RANGE, E_IO
read_score	study_id	score_result.json	E_NOT_FOUND, E_SCHEMA

2.7 Review package (US-01, I13–I15)

(a) ReviewPackage → review_package.json; package_status ∈ {built, viewer_open_failed, partial_no_score}.

(b) Still builds when score is null (pathologist sees localization). Overlay format OQ-02 (assumes GeoJSON + helper.json). No auto QuPath fallback (D0 D-08).

Name	Inputs	Outputs	Errors
build_review_package	study_id	review_package.json + review/ files	E_LOCALIZATION_UNRESOLVED, E_SCORE_FILE_MISSING, E_OVERLAY_WRITE, E_IO
mark_viewer_result	study_id, opened, note?	updated package_status	E_NOT_FOUND

2.8–2.11 Glue components

Authentication: No app-level auth. Hospital/UC SSO + folder ACLs. Observable error E_PERMISSION_DENIED; no retry.

Configuration: v1/config/pipeline_config.yaml — paths, identifier_rules, agreed_fields, scoring, logging. load_config / get_rule_set_version. Errors: E_CONFIG_NOT_FOUND, E_CONFIG_SCHEMA, E_CONFIG_PATH_INVALID. New I17.

Logging: Python logging → v1/logs/<component>/<run_id>.log. No PHI in messages (sanitizer). New I16. Retention OQ-03 (stub 90 days).

Notifications: None emitted. Status pulled via read_case_status (I18). Batch pipeline; no approved PHI-adjacent notify channel.

3. Edge cases and error handling

43 rows covering every component × boundary / timing / bad input / external dependency. R = recoverable, N = not.

ID	Component	Cat.	Edge case → decision	R/N
E-01	Gate	Bound	Zero-byte file → block E_UNKNOWN_FORMAT	R
E-04	Gate	Bad	Burned-in name in pixels → E_PIXEL_IDENTIFIER; not copied	N
E-05	Gate	Ext	Missing config → refuse start; no defaulted rules	R
E-09	Radiology	Ext	Cohort list unreachable → unmatched; never guess	R
E-11	Genomics	Time	Two workbooks same study_id → E_AMBIGUOUS_STUDY	R
E-12	Genomics	Bad	Non-xlsx types blocked at gate (D0 D-05)	N
E-14	Outcomes	Bound	Duplicate study_id rows → both ambiguous_join	R
E-16	Outcomes	Bad	Missing join column → E_MISSING_JOIN_COLUMN	R
E-19	Index	Time	Concurrent case_link writes → flock serialize	R
E-22	Integrator	Bound	Rad linked but annotation pending → insufficient unless molecular	R
E-25	Integrator	Ext	Pathology pointer missing → no input list; no U-Net retrain	N
E-28	Scorer	Bad	Feature missing → E_FEATURE_MISSING; no zero fabricate	N
E-29	Scorer	Ext	Model missing → block; review still partial_no_score	N
E-32	Review	Bad	Localization null → E_LOCALIZATION_UNRESOLVED	N
E-33	Review	Ext	PathPresenter reject → viewer_open_failed; no QuPath auto	R
E-36	Config	Bad	Unknown YAML keys → E_CONFIG_SCHEMA; refuse	R
E-40	Logging	Bad	Accidental PHI in message → [REDACTED] sanitizer	R
E-42	Auth	Ext	SSO expires → E_PERMISSION_DENIED; halt; restart after re-auth	R
E-43	Notify	Bad	Unknown study_id → E_NOT_FOUND; no empty fabricate	R

Full E-01…E-43 table is in D2_Detailed_Design.md §3 (every component × every category).

4. Requirements traceability

4.1 Forward: requirement → design elements

Req	Short text	Priority	Design elements (summary)
US-01	Pathologist sees localization helper	Must	§2.7 ReviewPackage; states InReviewPackage→OpenedInViewer; E-32/33
US-02	Risk estimate names only used inputs	Must	§2.5 inputs+decision; §2.6 used_inputs; partial-score rule; E-28
US-03	Radiology linked to one cohort case	Should	§2.3 link_status; I7; E-06/09
US-04	Outcomes join to exactly one case	Must	D1 ALG-01; OutcomesRow/ExclusionLog; E-14/16/17
US-05	Block/deid when identifier present	Must	§2.2 GateRecord/RedactionEntry; config rules; E-04/05
UC-01	Case risk estimate from inputs on hand	Must	§2.5–2.7; state machine §2.1
UC-02	Outcomes table model-ready	Must	D1 ALG-01 preserved
AC-01.1	One score + disclaimer	Must	ScoreResult schema
AC-01.2	No score if outcomes not ready	Must	outcomes_not_ready → block_reason
AC-01.3	No radiology → not in used_inputs	Must	Partial-score; molecular required if no rad
NFR-PHI	No PHI on laptops/repo	Must	Gate; log sanitizer; approved_root
NFR-COMPUTE	Approved compute only	Must	Config paths; model on share
NFR-RESEARCH	Research prototype disclaimer	Must	ScoreResult + ReviewPackage disclaimer
NFR-AUDIT	Trace config → score	Should	rule_set_version on outputs

4.2 Reverse (scope-creep check)

Every component maps to at least one requirement except Notifications, which exists only because Week 8 lists it as glue. It is scoped to “no notifications + read_case_status” so it is not scope creep.

5. Design rationale

ID	Chose X over Y because … accepted trade-off …
R-01	Deidentify with block fallback over block-only — normal CCHMC files have strip-able IDs; trade-off is maintaining identifier_rules + RedactionEntry audit (OQ-01).
R-02	Partial-score requires pathology+outcomes+(rad OR mol) over score-on-outcomes-alone — fused estimate is the project value; trade-off is fewer scorable cases near term.
R-03	Pre-annotated DICOM over in-pipeline feature extraction — annotation already upstream; trade-off is more pending cases.
R-04	Interface-first / model-TBD over picking model now — holdout comparison ongoing; trade-off is OQ-04 open; model_version keeps traceability.
R-05	Manual QuPath fallback over automatic switch — viewer used must be readable from output; human already in loop.
R-06	Pull status over push notifications — no approved PHI-adjacent channel; batch pipeline.
R-07	One YAML config over scattered env vars — faster audit; no secrets in file (SSO).
R-08	Config-driven 90-day log retention default over hard-code — OQ-03 institutional.
R-09	One read_case_status over six CLI tools — one surface for reviewers/pathologists.
R-10	Leave D0/D1 as historical record over rewriting — no peer-review errata yet; rule evolution stays visible.

6. Peer design review notes

Status: Scaffolded, not filled in. Peer review for Week 7 was not consolidated at writing time. After the session, replace with action items and update R-10 / team decisions if errata appear. Full consolidated review will live in Peer_Review_Notes.md.

ID	Criterion	Issue	Raised by	Action	Owner	Priority	Due
PR-01	to fill	to fill	team	to fill	Nipun/Shivam	to fill	to fill
PR-02							
PR-03							

Summary: criterion with most feedback / top two issues raised / must-fix applied — to fill.

Appendix A — Interface ID table

ID	From → to	Status
I1–I2	Sources → Identifier gate	D0; D2 refined (passed_with_redaction)
I3–I4	Gate → Radiology / Genomics intake	D0; D2 annotation_status on rad path
I5–I9	Outcomes / pathology / intakes → index	D0; I5/I9 detailed in D1
I10–I12	Index → Integrator → Scorer	D0; D2 assembly_decision enum
I13–I15	Scorer → Review → PathPresenter → Pathologist	D0; D2 partial_no_score
I16	Any component → Logging	New in D2
I17	Any component → Configuration	New in D2
I18	Any / human → read_case_status	New in D2

Appendix B — File / schema inventory under v1/

```
v1/
■■■ config/pipeline_config.yaml
■■■ incoming/<study_id>/<file> + <file>.gate_status.json
■■■ accepted/<study_id>/DICOM|xlsx/
■■■ cases/<study_id>/
■   ■■■ case_link.json, radiology.json, molecular.json, outcomes.json
■   ■■■ input_list.json, score_result.json, review_package.json
■   ■■■ review/<localization>.geojson, helper.json   (OQ-02)
■■■ cohort_study_ids.json
■■■ logs/<component>/<run_id>.log
```